Nathan Burns

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LinkedIn
Website

Education

University of Tennessee 2021 - 2027
Ph.D in Mathematics KNOXVILLE, TN
- *Thesis: The Free Boundary Curve Shortening Flow and Its Ancient Solutions*

University of Edinburgh 2016 - 2020
M.Math in Mathematics EDINBURGH, UK
- *First Class with Honours*

Research

Books

- T. Bourni, N. Burns, "Ancient Solutions to Free Boundary Curve Shortening Flow",
  Associação Instituto Nacional de Matemática Pura e Aplicada SEPTEMBER 2026
  ISBN: 978-85-244-0562-4

Publications

- T. Bourni, N. Burns, and M. Langford, "Stability of the Shrinking Semi-Circle Under the Free Boundary Curve Shortening flow",
  To Appear in: Communications in Analysis and Geometry ACCEPTED: AUGUST 2026
- T. Bourni, N. Burns and S. Catron, "Classification of convex ancient solutions to free boundary curve shortening flow in convex domains",
  The Journal of Geometric Analysis, Vol. 35, No. 7. JULY 2025

Preprints

- T. Bourni, N. Burns, and M. Flook, "Compact Mean Curvature Flows from Non-Compact Initial Data",
  arXiv: 2610.12817 (Submitted) OCTOBER 2026
- J. McAlister, N. Burns, "Myopic Best Response Replicates the Mean Curvature Flow"
  arXiv: 2609.37803 (Submitted) SEPTEMBER 2026

Miscellaneous Research

- A. Ho, S. Bagley, C. Beckford, N. Burns, E. Campbell, J. Kingsley and Mary E. Pilgrim, "In Between High School and College: A Qualitative Study on Educators' Self-Reported Teaching Practices",
  Journal of Education, DOI:10.1177/00220574261416889 JANUARY 2026
  - *This research was done as part of a research assistantship during the first year of graduate study.*

Service

Co-Organizer for Internal Allen-Cahn Reading Seminar 2026/2027 ACADEMIC YEAR
- University of Tennessee

Co-Organizer for the Geometric Analysis Seminar 2025/2026 ACADEMIC YEAR
- University of Tennessee

Academic Activity

Academic Visitor MARCH 2. - MARCH 8. 2025
- Australian National University

Program Associate OCT. 21. - NOV. 22. 2024
- SLMath (Formerly the MSRI)
- New Frontiers in Curvature: Flows, General Relativity, Minimal Submanifolds and Symmetry.

Talks

- Arizona State University - Geometry and Topology Seminar — OCTOBER 2026
- Auburn University - Topology and Geometry Seminar — SEPTEMBER 2026
- Johns Hopkins University - Geometric Analysis Seminar — APRIL 2026
- North Carolina State University - Geometry and Topology Seminar — APRIL 2026
- AMS Spring Southeastern Meeting: Geometric and Non-Smooth Analysis — MARCH 2026
- AMS Fall Central Meeting: Riemannian Geometry and Symmetry II — OCT. 2025
- Rutgers Gauge Theory, Low-Dimensional Topology, and Geometric Analysis — MAY 2025
- Australian National University - Geometric Analysis Seminar — MARCH 2025
- MATRIX-MFO Tandem Workshop: Nonlinear Geometric Diffusion Equations — FEB. 2025
- SLMath (Formerly the MSRI) - Graduate Student Seminar — NOV. 2024
- Punta Del Este (Uruguay) - Geometric Flows and Relativity Workshop — MARCH 2024

Conference Attendance

- AMS Spring Southeastern Meeting: Geometric and Non-Smooth Analysis — MARCH 2026
- Courant-Rutgers Geometric Analysis Workshop 2026 - Rutgers University — JAN. 2026
- Texas Geometry and Topology Conference - UT Dallas — NOV. 2025
- AMS Fall Central Meeting: AMS Special Session on Riemannian Geometry and Symmetry II — OCT. 2025
- Topics in Differential Geometry - Brown University — JUNE 2025
- Rutgers Gauge Theory, Low-Dimensional Topology, and Geometric Analysis — MAY 2025
- MATRIX-MFO Tandem Workshop: Nonlinear Geometric Diffusion Equations — FEB. 2025
- The 30th Southern California Geometric Analysis Seminar - UC Irvine — FEB. 2025
- Geometry and Analysis of Special Structures on Manifolds - SLMath — NOV. 2024
- Recent Progress on Geometric Analysis and Riemannian Geometry - SLMath — OCT. 2024
- Bridging PDE and Geometric Flows - Cardiff University — JULY 2024
- Kansas Geometric Analysis Conference - Wichita State University — APRIL 2024
- Geometric Flows and Relativity Workshop - Punta Del Este (Uruguay) — MARCH 2024
- Geometric Flows and Relativity Summer School - Universidad de La República — MARCH 2024

Teaching

All teaching as a graduate student at the University of Tennessee.

- **Instructor of Record**
  - Math 152 - Calculus for the Life Sciences (2 Sections)
  - Math 123 - Finite Mathematics (3 Sections)
  - Math 113 - Mathematical Reasoning (3 Sections)
- **Teaching Assistant**
  - Math 241 - Calculus III (1 Section)
  - Math 231 - Differential Equations I (1 Section)
  - Math 125 - Basic Calculus (1 Section)
  - Math 119 - College Algebra (1 Section)
  - Math 113 - Mathematical Reasoning (2 Sections)

Fellowships/Awards

All Awards Received as a Graduate Student at the University of Tennessee

- Summer Departmental Research Fellowship — SUMMER 2025
- Spike Tickle STEM Endowed Fellowship — SPRING 2025
- Graduate Student Fellowship: Passing UTK Oral Exam Early — SPRING 2024
- Graduate Student Academic Achievement Award — SPRING 2024
- Graduate Student Fellowship: Passing Preliminary Exams Early — SPRING 2023
- F.M Dryzer Memorial Fellowship — FALL 2021
- Graduate School Fellowship: Previous Outstanding Record — THROUGHOUT GRAD. SCHOOL